

Prasanjit Dubey (he/him)

Ph.D. Candidate, Machine Learning · Graduating May 2027

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Research Interests	Federated learning and knowledge distillation for language models, communication-constrained learning and inference, robustness to faulty and adversarial sources, evaluating and monitoring AI systems, anytime-valid inference, multiple hypothesis testing, conformal prediction.	

Education

Georgia Institute of Technology, Atlanta, GA, USA

Ph.D. in Machine Learning | GPA 3.85/4.00

Aug 2021 – May 2027 (expected)

- › H. Milton Stewart School of Industrial and Systems Engineering (ISyE), Minor in Mathematics
- › Advisor: Prof. Xiaoming Huo

Indian Institute of Technology Bombay, Mumbai, India

M.Sc. in Applied Statistics & Informatics | Class Rank 1 of 38 | GPA 8.96/10.00

2017 – 2019

- › Department of Mathematics
- › Thesis: *Financial Data Mining Using Multivariate Copulas*

University of Calcutta, Kolkata, India

B.Sc. in Statistics, First Class Honors | Grade 92.5%

2014 – 2017

- › Department of Statistics, Minor in Mathematics & Computer Science

Publications and Preprints

Author order is by contribution. * marks an award or a presentation associated with the paper.

Refereed Publications

- [1] **BOOST: Power-Optimal Strong-FWER Testing for Block-Structured Multiplicity.**
[Prasanjit Dubey](#) & Xiaoming Huo.
Conference on Neural Information Processing Systems (NeurIPS), 2026. [Paper]
- [2] **Generalization Bounds for Magnitude-Based Pruning via Sparse Matrix Sketching.**
Etash Kumar Guha, [Prasanjit Dubey](#), & Xiaoming Huo.
International Conference on Learning Representations (ICLR) Workshop on Bridging the Gap Between Practice and Theory in Deep Learning, 2024. [Paper]

Manuscripts Under Review and Preprints

- [3] **Pooling Sequential Evidence Across Hypotheses: Rate-Optimal Multiple Testing at a Fixed Horizon.**
[Prasanjit Dubey](#) & Xiaoming Huo.
Under review at the Electronic Journal of Statistics, 2026. [Paper]
- [4] **Report Resolution in Federated Multiple Testing under Family-Wise Error Control.**
[Prasanjit Dubey](#) & Xiaoming Huo.
Preprint, 2026. [Paper]
* Accepted at the Second Workshop on ML×OR: Mathematical Foundations and Operational Integration of Machine Learning for Uncertainty-Aware Decision-Making, NeurIPS 2026
- [5] **Byzantine-Robust Federated RAG via Aligned Calibration and Fixed-Membership Conformal Prediction.**
[Prasanjit Dubey](#), Aritra Guha, & Xiaoming Huo.
Under review at the International Conference on Learning Representations (ICLR), 2027.
- [6] **Matching Rates and Optimal Allocation for Federated Probe-Logit Distillation under Heterogeneous Bandwidth Budgets.**
[Prasanjit Dubey](#) & Xiaoming Huo.
Invited submission under review at the Journal of the American Statistical Association: Theory and Methods, 2026. [Paper]
* Poster, Statistics and Trustworthy AI for Cross (X)-Domain Acceleration (STAI-X), Harvard University, 2026

- [7] **Certified Adaptive Refresh: Anytime-Valid Monitoring for Federated Conformal RAG.**
 Prasanjit Dubey & Xiaoming Huo.
Under review at the International Conference on Learning Representations (ICLR), 2027. [Paper]
 * Accepted at the Second Workshop on ML×OR: Mathematical Foundations and Operational Integration of Machine Learning for Uncertainty-Aware Decision-Making, NeurIPS 2026
- [8] **Federated Language Models Under Bandwidth Budgets: Distillation Rates and Conformal Coverage.**
 Prasanjit Dubey & Xiaoming Huo.
Preprint, 2026. [Paper]
- [9] **Most Powerful Tests with Exact Family-Wise Error Rate Control: Necessary Conditions and Fast Computation for General K.**
 Prasanjit Dubey & Xiaoming Huo.
Revision under review at the Journal of Computational and Graphical Statistics (JCGS), 2025. [Paper]
[Companion paper] *Most Powerful Test with Exact Family-Wise Error Rate Control: Necessary Conditions and a Path to Fast Computing*
 * Invited talk, Optimisation for Statistics and Learning session, 2025 INFORMS Annual Meeting
- [10] **LLmFPCA-detect: LLM-powered Multivariate Functional PCA for Anomaly Detection in Sparse Longitudinal Texts.**
 Prasanjit Dubey, Aritra Guha, Zhengyi Zhou, Qiong Wu, Xiaoming Huo, & Paromita Dubey.
Preprint, 2025. [Paper]
 * 2nd Place Paper Award, Student Research Showcase, 2025 INFORMS Regional Analytics Conference (IRAC)

Invited Talks & Presentations

Matching Rates and Optimal Allocation for Federated Probe-Logit Distillation	
Poster, Statistics and Trustworthy AI for Cross (X)-Domain Acceleration (STAI-X), Harvard University	2026
Most Powerful Test with Exact Family-Wise Error Rate Control: Necessary Conditions and a Path to Fast Computing	
Invited talk, Optimisation for Statistics and Learning session, INFORMS Annual Meeting, Atlanta, GA	2025
Poster, Georgia Statistics Day, University of Georgia, Athens, GA	2025
LLmFPCA-detect: LLM-powered Multivariate Functional PCA for Anomaly Detection in Sparse Longitudinal Texts	
Poster, Student Research Showcase, INFORMS Regional Analytics Conference (IRAC), Atlanta, GA	2025
* 2nd Place Paper Award	

Awards & Honors

2nd Place Paper Award , Student Research Showcase, INFORMS Regional Analytics Conference (IRAC), Atlanta	2025
Stewart Topper Fellowship Award , Georgia Tech (first-year Ph.D. academic performance)	2021 – 2022
Dr. P. V. Sukhatme Memorial Award , IIT Bombay (M.Sc. rank 1)	2019
All India Rank 26 , Graduate Aptitude Test in Engineering (GATE), Statistics	2019
Pfizer Scholarship , IIT Bombay (M.Sc. first-year rank 1)	2018
All India Rank 7 , IIT Joint Admission Test for M.Sc. (IIT JAM), Mathematical Statistics	2017
Swami Asaktananda Award , University of Calcutta (highest B.Sc. academic distinction)	2017
Summer Research Fellowship , Indian Academy of Sciences	2016, 2017

Research & Professional Experience

Georgia Institute of Technology , Atlanta, GA, USA	Jan 2026 – Sep 2026
<i>Graduate Research Assistant, Georgia Department of Transportation Sponsored Research</i>	
Data Sufficiency at Continuous Count Stations · PI: Prof. Xiaoming Huo [Code]	
- Developed a statistical framework to establish how many valid days a traffic-monitoring station needs in a month with incomplete data before its monthly average can be published. - Applied the framework to the entire network of 233 traffic-monitoring stations across Georgia, delivering the results to transportation planners as the Data-Sufficiency Explorer, an interactive dashboard.	
Bloomberg L.P. , New York, USA	Jun 2025 – Aug 2025
<i>Quant PhD Intern, Quantitative Risk Analytics</i>	
- Built a Kalman-filter hazard-curve model for pricing credit instruments by estimating dynamic CDS term structures from latent Nelson–Siegel factors and macroeconomic covariates; integrated the model into Bloomberg Terminal's Credit VaR engine to forward-simulate risk-neutral hazard rates. - Implemented a Credit Rating Migration module for the Credit VaR engine using macro-conditioned multinomial	

logit and intensity-based models to simulate obligor- and index-level rating paths, capturing systematic and idiosyncratic risk.

AT&T Labs, Inc., New Jersey, USA

Research Scientist Intern, Data Science & AI Research

May 2024 – Aug 2024

- › Developed a novel framework for analyzing **sparse longitudinal customer-journey text** by integrating **large language models (LLMs)** with **multivariate functional data analysis**; work resulted in a manuscript (paper [10]).
- › Used **LLM-guided dynamic keyword profiling** on real-world data (Amazon reviews) to surface customer pain points and actionable insights, **outperforming existing methods for customer-journey analytics**; shipped a **production application** (Emotional AI) for behavioral anomaly detection and targeted interventions.

Australia and New Zealand Banking Group (ANZ), Bengaluru, India

Senior Quantitative Analyst, Wholesale Credit Risk Modeling

Jun 2019 – Aug 2021

- › Built the **Financial Ratio Surveillance Tool (FRST)**, a production **R Shiny** stress-testing application designed to forecast default probabilities and assess the predictive power of financial ratios.
- › Led the bank’s **statistical COVID-19 impact assessment** across commercial portfolios, flagging high-volatility sectors for stakeholders as part of **APRA regulatory** requirements.

Walmart Labs, Bengaluru, India

Research Intern, Data Science (International Pricing & Demand Forecasting)

May 2018 – Aug 2018

- › Devised an **ensemble of ridge, lasso, and elastic-net models** using high-dimensional, large-scale data to forecast sales across product categories based on item-, region-, and covariate-specific **price elasticities**.
- › Built a **cluster-based (k-means) price-elasticity model**, **reducing mean absolute percentage error (MAPE) by 42.4% relative to Walmart’s incumbent methods**.

Teaching Experience

Georgia Institute of Technology, Atlanta, GA, USA

Head Teaching Assistant and Teaching Assistant, H. Milton Stewart School of Industrial and Systems Engineering

2021 – 2025

- › **ISyE 7406, Data Mining and Statistical Learning** (graduate level, 300+ students)
Teaching Assistant: Fall 2024, Spring 2025, Fall 2025
Instructor: Prof. Xiaoming Huo
- › **ISyE 6414, Statistical Modeling and Regression Analysis** (graduate level, 300+ students)
Head Teaching Assistant: Fall 2022, Spring 2023 | **Teaching Assistant:** Summer 2022
Instructor: Prof. Nicoleta Serban
- › **ISyE 2027, Probability with Applications** (undergraduate level)
Student Assistant: Fall 2021 · Led tutoring sessions

Student Mentoring

- › **Kareena Mehta**, undergraduate, Georgia Institute of Technology (Computer Science) Summer 2026
- › **Kevin Li**, undergraduate, Washington University in St. Louis (Finance and Data Science) Summer 2026, Fall 2026

Professional Service

Conference Reviewer:

International Conference on Learning Representations (ICLR)
International Conference on Machine Learning (ICML)
Conference on Neural Information Processing Systems (NeurIPS)

2023 – present

Technical Skills

Programming:	Python, R, R Shiny, C++, SQL, MySQL, MATLAB
Machine Learning:	LLMs, PyTorch, scikit-learn
Computing & Tools:	High-performance computing (Slurm), GPU computing, Apache Hadoop (MapReduce, Streaming), Git, LaTeX
Relevant Graduate Coursework:	Mathematical Foundations of Machine Learning, High-Dimensional Probability, High-Dimensional Statistics, Statistical Inference, Time Series Analysis, Stochastic Processes, Convex Optimization, Stochastic Optimization, Computational Data Analysis